**(i)**

$\because e^x$ is increasing,

$$\therefore P(Y \leq e^{x_m}) = P(e^X \leq e^{x_m}) = P(X \leq x_m) = \frac{1}{2}$$

$$\therefore \boxed{y_m = e^{x_m}}$$

---

**(ii)**

$$P(Y \leq y) = P(e^X \leq y) = P(X \leq \ln y) = F(\ln y)$$

$$\implies g(y) = \frac{d}{dy} F(\ln y) = f(\ln y) \cdot \frac{1}{y}$$

When $y = \lambda \implies g'(\lambda) = 0$

$$g'(y) = \frac{d}{dy}\left(\frac{f(\ln y)}{y}\right) = \frac{f'(\ln y) - f(\ln y)}{y^2}$$

$$\implies f'(\ln \lambda) - f(\ln \lambda) = 0$$

---

**(iii)** Since the sum of $f(x) = 1$

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}\, dx = 1$$

When $x = v - \sigma^2 \implies \dfrac{1}{\sigma\sqrt{2\pi}} \displaystyle\int_{-\infty}^{\infty} e^{-(v-\mu-\sigma^2)^2/(2\sigma^2)} dv = 1$

$$\mathrm{E}(Y) = \mathrm{E}(e^X) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^x e^{-(x-\mu)^2/(2\sigma^2)}\, dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{x-(x-\mu)^2/(2\sigma^2)}\, dx$$

$$x - \frac{(x-\mu)^2}{2\sigma^2} = \mu + \frac{1}{2}\sigma^2 - \frac{(x-\mu-\sigma^2)^2}{2\sigma^2}$$

$$\implies \mathrm{E}(Y) = e^{\mu+\frac{1}{2}\sigma^2} \cdot \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-\mu-\sigma^2)^2/(2\sigma^2)}\, dx = \boxed{e^{\mu+\frac{1}{2}\sigma^2}}$$

---

**(iv)** From part (i), $x_m = \mu \implies y_m = e^{\mu}$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)} \implies f'(x) = -\frac{x-\mu}{\sigma^2} f(x)$$

From part (ii), $f'(\ln \lambda) = f(\ln \lambda)$

$$-\frac{\ln \lambda - \mu}{\sigma^2} f(\ln \lambda) = f(\ln \lambda)$$

$$\implies \ln\lambda = \mu - \sigma^2 \implies \lambda = e^{\mu-\sigma^2}$$

$$\because \mathrm{E}(Y) = e^{\mu+\frac{1}{2}\sigma^2}$$

Since $\sigma^2 > 0$,

$$\therefore e^{\mu-\sigma^2} < e^{\mu} < e^{\mu+\frac{1}{2}\sigma^2}$$

$$\boxed{\lambda < y_m < \mathrm{E}(Y)}$$

---

This solution was submitted by **Jon**
Published and refined by **Jon**

*Last updated: June 24, 2026*